Infinite Sums at the Boundary: Uniform Convergence, Analyticity, and Abel Limits

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The same infinite series can converge at one point and diverge at another. Even where it converges pointwise, differentiation or integration may fail to commute with summation. Power series are unusually well behaved inside their convergence disks, while interesting phenomena accumulate on the boundary. Abel's theorem connects these two regimes by approaching the boundary from inside, and an oscillatory integral gives a continuous analogue of the same regularization.[1]

§1.1 Pointwise convergence is too weak for calculus

The convergence of $\sum_{n\geq 1} a_n$ means that its partial sums have a finite limit; $a_n \to 0$ is necessary but not sufficient. The harmonic series diverges because

$$\frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots$$

contains infinitely many blocks of sum at least $1/2$.

For a series of functions $\sum_n f_n(x)$, uniform convergence means

$$\sup_{x\in X} \left| \sum_{n=1}^{N} f_n(x) - f(x) \right| \longrightarrow 0.$$

The Weierstrass M-test supplies a practical criterion: if $|f_n(x)| \leq M_n$ on X and $\sum M_n < \infty$, then $\sum f_n$ converges uniformly and absolutely.

Uniform convergence of continuous functions preserves continuity. On a compact interval, it also allows integration term by term:

$$\int_a^b \sum_{n=1}^{\infty} f_n(x)\, dx = \sum_{n=1}^{\infty} \int_a^b f_n(x)\, dx.$$

Differentiation requires stronger information, typically uniform convergence of the derivative series together with convergence at one reference point.

An elementary contrast is $f_n(x) = x^n$ on $[0, 1]$. It converges pointwise to zero for $x < 1$ but stays equal to 1 at $x = 1$. The limit is discontinuous, so the convergence cannot be uniform. The phenomenon occurs at the boundary where all the smallness estimates break down simultaneously.

§1.2 Power series are analytic inside a disk

Consider

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n.$$

[1] For complex power series and analytic continuation, see MIT 18.04, *Complex Variables with Applications*; for Abelian summation methods, see G. H. Hardy, *Divergent Series*. https://ocw.mit.edu/courses/18-04-complex-variables-with-applications-spring-2018/.

Its radius of convergence is determined by

$$R^{-1} = \limsup_{n\to\infty} |a_n|^{1/n},$$

with the usual conventions for $R = 0$ and $R = \infty$. For every $r < R$, the root test provides a summable majorant for $|a_n|r^n$; hence the series converges uniformly on the closed disk $|z - z_0| \le r$.

The differentiated series

$$f'(z) = \sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1}$$

has the same convergence radius, because $n^{1/n} \to 1$. Thus a complex power series can be differentiated termwise any number of times inside its disk. In particular, its sum is holomorphic there.

The real function

$$f(x) = \frac{1}{1 + x^2}$$

is smooth on the entire real line, but its Taylor series at zero is

$$f(x) = 1 - x^2 + x^4 - x^6 + \cdots, \qquad |x| < 1.$$

The obstruction at radius 1 is invisible on the real axis: as a complex function, $1/(1+z^2)$ has poles at $z = \pm i$. This explains why real smoothness is insufficient to guarantee a large Taylor radius. Complex singularities govern the analytic power series.

§1.3 Abel's theorem recovers an ordinary sum from inside

Suppose $\sum_{n=0}^{\infty} a_n = S$, and write

$$s_n = \sum_{k=0}^{n} a_k.$$

For $0 < r < 1$, summation by parts gives

$$\sum_{n=0}^{\infty} a_n r^n = (1 - r) \sum_{n=0}^{\infty} s_n r^n.$$

The right-hand series converges because the convergent partial sums s_n are bounded. Since $s_n \to S$,

$$\begin{aligned}(1 - r) \sum_{n=0}^{\infty} s_n r^n - S &= (1 - r) \sum_{n=0}^{\infty} (s_n - S) r^n \\ &\longrightarrow 0 \qquad (r \uparrow 1).\end{aligned}$$

To justify the last step, split the sum at a fixed N. The finitely many early terms are multiplied by $1 - r$ and vanish; the remaining terms are bounded by $\sup_{n\ge N} |s_n - S|$, which can be made arbitrarily small.

We have proved Abel's theorem:

$$\boxed{\sum_{n=0}^{\infty} a_n = S \implies \lim_{r\uparrow 1} \sum_{n=0}^{\infty} a_n r^n = S.}$$

The factors r^n damp the distant terms, turning a boundary sum into a family of absolutely convergent interior sums.

§1.4 The converse needs more information

The series $1 - 1 + 1 - 1 + \cdots$ does not converge in the ordinary sense; its partial sums alternate between 1 and 0. Yet for $0 < r < 1$,

$$1 - r + r^2 - r^3 + \cdots = \frac{1}{1+r} \longrightarrow \frac{1}{2} \qquad (r \uparrow 1).$$

The value $1/2$ is its *Abel sum*, not its usual sum. Abel summability is strictly more permissive than ordinary convergence.

For nonnegative terms $a_n \geq 0$, however, a finite Abel limit does force convergence. Indeed

$$\sum_{n=0}^{\infty} a_n r^n \uparrow \sum_{n=0}^{\infty} a_n \quad (r \uparrow 1),$$

by monotone convergence applied to the counting measure on $\mathbb{N}$. Therefore, if the left side tends to $L < \infty$, then the ordinary series converges and equals L.

This is a simple Tauberian principle: the positivity of the coefficients prevents oscillatory cancellation from concealing divergence. Stronger Tauberian theorems replace positivity by quantitative assumptions on the decay or sign changes of coefficients.

§1.5 Oscillatory improper integrals behave similarly

The improper integral

$$\int_1^{\infty} \frac{\sin x}{x}\, dx$$

converges by integration by parts:

$$\int_A^B \frac{\sin x}{x}\, dx = \left[-\frac{\cos x}{x}\right]_A^B - \int_A^B \frac{\cos x}{x^2}\, dx.$$

The boundary term converges as $B \to \infty$, and the remaining integral is absolutely convergent. But the original integral is not absolutely convergent: over successive half-periods the integral of $|\sin x|/x$ is bounded below by a constant multiple of $1/k$, giving a divergent harmonic comparison.

For $t > 0$, introduce exponential damping:

$$F_t(a) = \int_0^{\infty} e^{-tx} \frac{\sin(ax)}{x}\, dx.$$

Differentiating with respect to a, justified by the integrable majorant e^{-tx}, gives

$$F_t'(a) = \int_0^{\infty} e^{-tx} \cos(ax)\, dx = \frac{t}{t^2 + a^2}.$$

Since $F_t(0) = 0$, integration in a yields

$$\boxed{F_t(a) = \arctan \frac{a}{t}.}$$

At $a = 1$, the limit $t \downarrow 0$ is $\pi/2$. To pass to the undamped integral, observe that for every $A > 0$ and $t \geq 0$, integration by parts together with the positivity and monotonicity of e^{-tx}/x gives

$$\left| \int_A^{\infty} e^{-tx} \frac{\sin x}{x}\, dx \right| \leq \frac{2}{A}.$$

The tail is therefore uniformly small for all $t \geq 0$. On the bounded interval $[0, A]$, dominated convergence permits $t \downarrow 0$ under the integral. Combining the two bounds gives

$$\lim_{t \downarrow 0} F_t(1) = \int_0^\infty \frac{\sin x}{x}\, dx = \boxed{\frac{\pi}{2}}.$$

For comparison, the simpler oscillatory integral $\int_0^\infty \sin x\, dx$ does not converge as an ordinary improper integral, even though

$$\int_0^\infty e^{-tx} \sin x\, dx = \frac{1}{1+t^2} \longrightarrow 1.$$

The regularized value 1 is not an ordinary integral. As with $1 - 1 + 1 - \cdots$, damping can create a boundary value without making the undamped object convergent.

Power series, Abel limits and oscillatory integrals are governed by the same issue: operations performed safely in an interior convergence regime may require additional information at the boundary. Uniform estimates justify passage to a limit, positivity can prevent hidden cancellation, and complex analyticity reveals obstructions that a purely real calculation may overlook.